

Spring 2026
Introduction to Philosophy of Technology
Quiz-2
31st March, 2026

Time: 45 minutes

Marks: 20 (weighted to 10)

Answer any 2 of the following: (2 × 10; 300 words)

1. What is a theory in science? What is the 'model = theory' myth in computer science? Why is it dubious to think that programs are theories? What implications does this have for computer science as a science?

Theory in science (3; any 3 points; 1 mark each)

- In science in general, the term theory is sometimes used for **relating laws with principles and ideas different from those used in the law itself**.
- For instance, the Kinetic **Theory** of Gases is a framework that **explains Boyle's Law** by invoking an idea of the **atomic nature of gas as "billiard balls"** flying and bouncing about.
- A set of **laws used to explain and predict a set of events**.
- **One can have a theory without having a model**. For instance, one can have quantum theory without having a model of quantum mechanics.
- **one can have a model of a subject matter without having a theory** about that subject matter. Soap film example.

Model=theory myth (1 mark)

- In computer science, the "model = theory myth" states that **constructing a computer model of a phenomenon is the same as constructing a theory about that phenomenon**.

Why it's dubious to think that programs are theories (4)

Examples where programs are thought of as theories (2 marks; any two examples; 1 mark each)

- **Example from AI**
 - there is a common confusion between programs and theories. Generally in Artificial Intelligence, when someone asks for a specification of the theory behind a program, **the response usually given is that the program is the theory**".
- **Example from Newell and Simon**
 - who equated "theory of thinking" with the ability to simulate human problem solving behavior.
- **Example of ELIZA program**
 - the ELIZA program is a theory of human conversation (or of thinking, or of the mind) or
- **Example of Kids' club's football playing robots**
 - whether Kids' Club's football playing robots run the theory of playing football
- **Or any other example along similar lines**

Why it's dubious (2 marks; 1 mark each; look for any 2 points)

- Since one can have a model without having a theory about the subject matter, **mere presence of a model or a program does not entail the presence of a theory behind the model.**
- The **ability to model a phenomenon** does **not necessarily mean that one would understand the phenomenon.**

Implications for computer science as a science (2 marks; any 1 point)

- It is not sure what is being tested in running a program successfully. If the model=theory myth is valid, then in successfully running a program there is **no testing of theory**. Hence, computer science is **not analogous to the scientific method of testing hypothesis through experiments.**
 - Another analogy between the scientific method and problem solving in computer science is that **“programming is to computer science what the laboratory is to the physical sciences”**. But if the model=theory myth is valid, then there is no theory that is being tested by experimentation through programming.
 - Or any other valid point. Use your discretion here.
-

2. Briefly outline the critique of technology as applied science from
 - (a) philosophy of science
 - (b) people's history of science

Critique of technology as applied science from (a) philosophy of science (5 marks; 1 marks for each point; look for any 5)

- Recent approaches to the philosophy of science have shown that **science is laden with philosophical presuppositions**, and many feminists, ecologists, and other social critics of science have claimed that **science also is laden with social presuppositions.**
- The technology as applied science conception **ignores the knowledge component of technology**. In thinking of science as producing knowledge and technology as the application of this knowledge, it does not allow for technology to have a knowledge component. However, there is realm of know-how or skill that cannot be reduced to know-that and that can exist independently of scientific knowledge or theory.
- There is a **component of skill involved in doing science**. Hence, there is a **techne or a know-how involved in the activity of science** which makes it out to be more similar to technology. Hence
- The **instrumental realists developed a strong focus on the embodied**, active aspects of science that became a significant movement within the philosophy of science. For the instrumental realists **scientific instrumentation is central to science**. The **active, manipulative aspect of instrumental observation takes priority over passive observation and contemplation.**
- Since contemporary science is so involved with and dependent upon sophisticated technological instrumentation, **insofar as scientific discovery is based on observation, technology is prior to science as well as driving science.**
- The positivist focus on science as objective and value-neutral conceals how science **not only represents the world but also intervenes into the world**. Scientific theories and models, like technology, shape not just our beliefs but also the way we act and perceive in the world. In this sense, science and technology are more intertwined than the linear model (technology as applied science) suggests.

Critique of technology as applied science from (a) people's history of science (5 marks; 5 marks for each point; any 5)

- the idea that technology is applied science is based on the **facile assumption that scientific theory has always been a precondition of technological advance**
 - From a people's history of science, it is **technology that has driven the growth of scientific knowledge.**
 - The beginning of science is **not the word but the deed—not the proclamations of brilliant theorists but the creative handiwork of ordinary people.**
 - Until the last century or two, the process of gaining knowledge of nature has generally been **more a product of hands than brains; that is, of empirical trial-and-error procedures rather than theoretical application.**
 - Rigidly separating the histories of science and technology serves to reinforce the fallacious notion that **science arose from the realm of pure thought, floating in the clouds above the world of mundane human pursuits.** A People's history of science shows how science originated in, and was at first identical with, the practical crafts.
 - The **historical priority of technology over theoretical science** is most generally exemplified by the **central theme of this book**, which is that **artisans contributed not only the mass of empirical knowledge that furnished the raw material of the Scientific Revolution, but the empirical method itself.**
 - **Look also for any examples to illustrate this point.**
-

3. What are the metaphors with which the realist and instrumentalist views of science work? Briefly outline the thesis which describes the Renaissance birth of modern science as the coming together of these two metaphors.

Metaphors with which the realist and instrumentalist views of science work

Realist views of science (3; any 2; 1.5 marks each)

- Realists often describe scientific theories as “**pictures**” of the world.
- Scientific realists claim that the theoretical terms in science **represent or refer to objectively real entities, even if we cannot observe them.**

Instrumentalist views of science (3; any 2; 1.5 marks each)

- Instrumentalists describe theories as “**tools**” for prediction.
 - Or instrumentalists treat theories merely as **instruments for prediction**
- The **theories do not describe real, unobserved, structures, but are more or less useful for prediction of things we can directly observe.**

The thesis which describes the Renaissance birth of modern science as the coming together of these two metaphors (4 marks; 1 mark for getting the name ‘Zilsel thesis’ right; 1 mark for each of the points; any 3; of which the first point is mandatory; other 2 could be any of the others)

- The claim that the **breakdown of class barriers led to communication between humanists and artisans (Zilsel, 2000)** is called **“the Zilsel thesis.”** (MANDATORY POINT!)
- The birth of early modern science may have been the **fusion of literate scholars**, knowledgeable of the Greek classics and philosophical “world pictures,” but ignorant of practical crafts, **with the illiterate but technologically skilled artisans with their tools.**
- **Economic hard times** in the Renaissance **threw together impoverished wandering scholars with wandering artisans**, yielding the **“marriage of metaphysics and technology”** that is science.
- The **“tools” in the form of painter’s brush or sculptor’s chisel** were a means to the **“picture” in the form of Renaissance painting or sculpture.** These components of the birth of modern science and technology show up today in the preferred metaphors of instrumentalism and realism as philosophies of the relation of science to reality.